Question	Answer	Marks	Guidance
2(a)(i)	force per unit mass (at the point)	**B1**	Ratio must be clear. **Ignore** symbols unless defined. **Ignore** references to units.
2(a)(ii)	four straight parallel lines perpendicular to ground approximately evenly spaced	**B1**	**Allow** more than four lines. **Allow** tolerance ±5° to the vertical. Lines do not need to reach the Earth's surface. Judge even spacing only in relative to the other lines. Assess spacing by eye; **Do not allow** only if the spacing is so uneven that there is no need to check with a ruler.
	lines pointing towards the ground	**B1**	Minimum one arrow needed, but all arrows shown must be towards the ground.
2(b)(i)	$g = GM / R^2$	**C1**	
	mass = $9.81 \times (6.37 \times 10^6)^2 / (6.67 \times 10^{-11})$ $= 5.97 \times 10^{24}$ kg	**A1**	Must be given to 3 significant figures. AFC.
2(b)(ii)	$\phi = -(6.67 \times 10^{-11} \times 5.97 \times 10^{24}) / (6.37 \times 10^6)$ **or** $\phi = -9.81 \times 6.37 \times 10^6$	**C1**	**Allow** the C1 mark without the minus sign. Possible ECF from **2(b)(i)** for first method.
	$\phi = -6.25 \times 10^7$ J kg^{-1}	**A1**	Must be given to 3 significant figures. AFC. Answer must be negative for the A1 mark. Unit needed with answer for the A1 mark. **Allow** N m kg^{-1} or m^2 s^{-2} for J kg^{-1}.

Question	Answer	Marks	Guidance
2(b)(iii)	GPE = potential × mass = (–) GMm/R $= -(6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 96.0) / ((6.37 + 1.93) \times 10^{6})$	C1	**Allow** the C1 mark without the minus sign. Possible ECF from **2(b)(i)**. Use of only 6.37 or only 1.93 in radius of orbit is XP in the C1 mark.
	$= -4.61 \times 10^{9}$ J	A1	Correct to at least 2 significant figures. AFC. (4.61) Treat AFC answer as -4.6×10^{9} J. Must be negative unless lack of minus sign was the direct cause of the A1 mark not being awarded in **2(b)(ii)**.